Microprocessors and Microcontrollers

Assignment 9- Week 9

TYPE OF QUESTION: MCQ

Number of questions: 15 **Total mark: 15 X 1 = 15 M**

QUESTION 1:

Correct Answer: c

Detailed Solution:

The ARM10TDMI processor is based on a 6-stage pipeline, which improves performance and instruction throughput compared to its predecessors. The six pipeline stages are:

1. Fetch (F) – The instruction is fetched from memory.
2. Decode (D) – The fetched instruction is decoded.
3. Register Read (R) – The required registers are read.
4. Execute (E) – The ALU or other execution units process the instruction.
5. Memory Access (M) – Accesses memory if needed (for load/store operations).
6. Write-back (WB) – Writes the result back to the register file.

Q2

In an ARM-based processor with a multi-stage pipeline, which of the following is true regarding bus architecture?

a) Both instruction and data buses use the same width
b) The instruction bus is narrower than the data bus for performance optimization
c) The data bus is narrower than the instruction bus for memory efficiency
d) The instruction and data buses have dynamically adjustable widths

Ans: (b)

Explanation:

Q3.

Ans: (c)

Q4.

Ans: (d)

Explanation:

- ARM supports both single and multiple load/store instructions.
- The LDM (Load Multiple) and STM (Store Multiple) instructions allow transferring multiple registers in a single instruction, significantly improving performance.
- The ARM architecture allows loading/storing up to 16 registers at once using LDM/STM instructions.
- This is beneficial for fast context switching, function calls, and handling large data efficiently.

Q5.

Ans: (b)

Explanation:

- In the ARM architecture, when a procedure (subroutine) call is made using the BL (Branch with Link) instruction, the return address (the address of the next instruction after the call) is automatically stored in R14, which is known as the Link Register (LR).
- This allows the processor to return to the correct instruction after the subroutine finishes by simply copying the value of R14 into the Program Counter (R15) using the MOV PC, LR instruction.
- The Stack Pointer (R13) is used for managing the stack but is not automatically used for storing the return address unless explicitly done by the programmer.

Q6.

Ans: (c)

Explanation:

- In ARM architecture, a Software Interrupt (SWI) instruction is used to switch from User mode to Supervisor mode (SVC), which is a privileged mode.
- This is commonly used for system calls, where a user application requests services from the operating system (like accessing hardware or memory management).
- FIQ (Fast Interrupt) and IRQ (Normal Interrupt) are used for hardware-generated interrupts, not software interrupts.
- Abort mode is entered when there is a memory access violation or instruction fetch error.

Q7.

Ans: (c)

Explanation:

- CPSR (Current Program Status Register) holds important processor state information, such as condition flags, mode bits, and interrupt disable flags.
- In privileged modes (FIQ, IRQ, SVC, etc.), CPSR can be modified by the processor or software.
- User mode is non-privileged, meaning it cannot modify CPSR directly.
- To modify CPSR in User mode, a software interrupt (SWI) or an exception must be used to switch to a privileged mode.

Q8.

Ans: (c)

Explanation:

- When an ARM processor resets, it automatically enters Supervisor Mode (SVC).
- Supervisor Mode (SVC) is a privileged mode that allows the processor to initialize system settings, configure registers, and set up memory before switching to User Mode or another mode.
- In this mode, the processor has full control over system operations and can execute privileged instructions.

Q9.

Ans: (d)

Explanation:

- Multiplying a value by **9** can be rewritten using bitwise shifts and addition:

$$r0 \times 9 = r0 \times (8 + 1) = (r0 \times 8) + r0$$

- In ARM assembly, left shifting a number by **3** (`LSL #3`) is the same as multiplying it by **8**:

$$r0 \times 8 = r0 \ll 3$$

- So, the equivalent ARM instruction to perform `r0 * 9` is:

$$r7 = r0 + (r0 \ll 3)$$

Q10.

Ans: (d)

Explanation:

- Multiply-Accumulate (MAC) instructions perform multiplication and add the result to an accumulator register.
- MUL → Performs simple multiplication but does not accumulate.
- UMULL → Unsigned Multiply Long, but it does not accumulate.
- SMULL → Signed Multiply Long, but again, it does not accumulate.
- SMLAL → Signed Multiply-Accumulate Long, which performs multiplication and adds the result to an accumulator register.

Q11.

Ans: (c)

Explanation:

- LDR → Loads a full 32-bit word (not specific to half-word).
- LDRS → This is not a valid ARM instruction.
- LDRH → Loads an unsigned half-word (16-bit) into the register.
- LDRSH → Loads a signed half-word (16-bit) and sign-extends it to 32-bit, which matches the question's requirement.

Q12.

Ans: (d)

Explanation:

- ARM multiplication instructions (`MUL`, `MLA`, etc.) require general-purpose registers for operands and result storage.
- R0, R1, and R14 can be used in multiplication instructions.
- R15 (PC - Program Counter) is not allowed because it controls instruction flow and using it in multiplication would disrupt execution.

Q13.

Ans: (b)

Explanation:

- Booth’s algorithm is widely used in ARM processors for multiplication operations, especially for handling signed numbers efficiently.
- It reduces the number of required addition and subtraction operations, making it more efficient than simple array multipliers.
- ARM uses Booth's and modified Booth's algorithms for high-speed multiplication in DSP (Digital Signal Processing) and general computing tasks.

Q14.

For the instruction below, which of the following statement is TRUE?

EQADD R0, R1, R2

a) R0 = R1 + R2 Only if parity flag is set
b) R0 = R1 + R2 Only if zero flag is set
c) R0 = R1 + R2 Only if carry flag is set
d) R0 = R1 + R2 Only if zero flag is reset

Ans: (B)

Explanation:

- The EQADD instruction in ARM stands for "Equal Add", which is a conditional execution of the ADD instruction.
- The EQ (Equal) condition is based on the Z (Zero) flag in the CPSR (Current Program Status Register).
- If Z = 1 (meaning the previous operation resulted in zero), then the instruction performs the addition: R0=R1+R2
- If Z ≠ 1 (meaning the result of the previous operation was not zero), the instruction is skipped.

Q15.

Ans: (b)

Explanation:

- In ARM instruction encoding, bits 31 to 28 store the condition code (COND field).
- ARM supports 16 condition codes, which determine whether the instruction executes based on flag conditions in the CPSR (Current Program Status Register).
- The condition codes allow conditional execution without the need for explicit branching.

Bit Allocation in a Standard ARM Instruction (32-bit format)

Bits	Field Name	Description
31-28	Condition Code (COND)	Specifies execution conditions based on CPSR flags
27-25	Instruction Class	Defines the instruction type (Data Processing, Load/Store, etc.)
24-21	Opcode	Specifies the operation (ADD, SUB, MOV, etc.)
20	Set Condition Flags (S-bit)	Determines if CPSR flags are updated

Bits	Field Name	Description
19-0	Operand fields	Varies by instruction type

Thus, the condition field (bits 31-28) allows ARM instructions to be conditionally executed.

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